



BHABHA UNIVERSITY ,BHOPAL

Scheme of Examination

Sixth Semester- Bachelor of Technology (Electrical Engineering) Session 2020-21

SPECIALIZATION – Electrical Engineering

S.No.	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory Slot)			Maximum Marks (Practical Slot)		Total Marks
			L	T	P		End Sem. Exam	Mid sem Tests	Assignments /Quiz	End Sem. Practical /Viva	Practical Record/ assignment/Quiz /Presentation	
1	EE-61	Power System-II	2	1	2	4	70	20	10	30	20	150
2	EE-62	Energy conservation management	3	1		4	70	20	10			100
3	EE-63	Power System Protection	3	1	2	5	70	20	10	30	20	150
4	EE-64	High voltage Engineering	3	1		4	70	20	10	-	-	100
5	EE-65	Electrical Simulation Lab			4	2	-	-	-	30	20	50
6		Industrial Training	To be completed anytime during Fifth/Sixth semester. Its evaluation/credit to be added in Seventh Semester									
7	EE-66	Minor Project II	-	-	6	3	-	-	-	-	100	100
8	EE- 67	Seminar			4	2					100	100
		Total	11	04	18	24	280	80	40	90	260	750

Theory Credits : 15
Practical Credits : 09
Total Credits : 24

Theory Marks : 280
Practical Marks : 90
Quiz, Mid Term, Lab. Work : 380

Total : 750



BHABHA UNIVERSITY, BHOPAL

**COURSE CURRICULUM
FACULTY OF ENGINEERING**

Undergraduate Programme

(Courses effective from Academic Year 2020-21)

**SYLLABUS OF COURSES TO BE OFFERED
“Bachelor of Technology”**

Disclaimer: The Undergraduate programme in Bachelor of Technology syllabus & schemes is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

Dean faculty of Engineering, Bhabha University

NH-12, Jatkhedi, Hoshangabad Road, Bhopal (M.P.) – 462 026

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Bachelor of Technology
VI
Semester
Electrical Engineering (EE)

NH-12, Jatkhedi, Hoshangabad Road, Bhopal (M.P.) – 462 026

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Bachelor of Technology

Branch - Electrical Engineering

EE-61 Power System-II

Unit-I

Power Flow Analysis :-

Review of the structure of a Power System and its components. Analysis of Power Flows: Formation of Bus Admittance Matrix. Real and reactive power balance equations at a node. Load and Generator Specifications. Application of numerical methods for solution of nonlinear algebraic equations – Gauss Seidel and NewtonRaphson methods for the solution of the power flow equations. Computational Issues in Large-scale Power Systems.

Unit-II

Stability Constraints in synchronous grids :-

Swing Equations of a synchronous machine connected to an infinite bus. Power angle curve. Description of the phenomena of loss of synchronism in a single-machine infinite bus system following a disturbance like a three--phase fault. Analysis using numerical integration of swing equations (using methods like Forward Euler, RungeKutta 4th order methods), as well as the Equal Area Criterion. Impact of stability constraints on Power System Operation. Effect of generation rescheduling and series compensation of transmission lines on stability.

Unit-III

Control of Frequency and Voltage :-

Turbines and Speed-Governors, Frequency dependence of loads, Droop Control and Power Sharing. Automatic Generation Control. Generation and absorption of reactive power by various components of a Power System. Excitation System Control in synchronous generators, Automatic Voltage Regulators. Shunt Compensators, Static VAR compensators and STATCOMs. Tap Changing Transformers. Power flow control using embedded dc links, phase shifters.

Unit-IV

Monitoring and Control:-

Overview of Energy Control Centre Functions: SCADA systems. Phasor Measurement Units and Wide-Area Measurement Systems. State-estimation. System Security Assessment. Normal, Alert, Emergency, Extremis states of a Power System. Contingency Analysis. Preventive Control and Emergency Control.

Unit-V

Power System Economics and Management :-

Basic Pricing Principles: Generator Cost Curves, Utility Functions, Power Exchanges, Spot Pricing. Electricity Market Models (Vertically Integrated, Purchasing Agency, Whole-sale competition, Retail Competition), Demand Side-management, Transmission and Distributions charges, Ancillary Services. Regulatory framework.

Bachelor of Technology

Branch - Electrical Engineering

REFERENCE BOOKS:-

1. Protection & Switchgear, B. Bhalja, R.P. Maheshwari, N.G.Chothani, Oxford.
2. Power system protection & switchgear, B.Ram & D.N. Vishwakarma, Tata McGraw Hill.
3. Handbook of Electrical Power Distribution, G. Ramamurthy, University Press
4. Electric Power Transmission and Distribution, S. Sivanagaraju, S.Satyanarayana, Pearson Education.
5. Power Systems Stability, Vol. I,II & II, E.W. Kimbark, Wiley.
6. Power Engineering, D.P Kothari & I.J. Nagrath, Tata McGraw Hill.
7. Power Systems Analysis, A. R. Bergen & V. Vittal, Pearson Education.
8. Computer Aided Power systems analysis, Dr. G. Kusic, CEC press.

EXPERIMENT LIST:-

1. Study on (i) on load Time Delay Relay (ii) off load Time Delay Relay.
2. Polarity, Ratio and Magnetisation Characteristics Test of CT & PT.
3. To develop a program in Matlab for information of Y-bus matrix for N bus system.
4. Load flow solution for 3-bus system using Gauss Seidel, Newton Raphson and FDLF methods up to 3 iteration.
5. Load flow solution for IEEE 6-bus and 30-bus system in Matlab using Newton Raphson method.
6. Assessment of transient stability of a single machine system.
7. Effect of compensation on voltage profile of IEEE 6-bussystem.
8. Study of any software tools (PSAT, EDSA, MY POWER, ETAP etc).

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Branch - Electrical Engineering

EE-62 Energy Conservation and Management

Unit-I

Introduction to energy & power scenario of world, National Energy consumption data, environmental aspects associated with energy utilization; Energy Auditing- need, types, methodology and barriers, role of energy managers, instruments of energy auditing.

Unit-II

Components of EB billing, HT and LT supply, transformers, cable sizing; Concept of capacitors, power factor improvement, harmonics; Electric motors- motor efficiency computation, energy efficient motors; Illumination Lux, Lumens, types of lighting, efficacy, LED lighting and scope of energy conservation in lighting.

Unit-III

Thermal systems, Boilers, Furnaces and Thermic Fluid heaters- efficiency computation and energy conservation measures; Steam distribution and usage, steam traps, condensate recovery, flash steam utilization; Insulation & Refractories.

Unit-IV

Energy conservation in major utilities; pumps, fans, blowers, compressed air systems, Refrigeration & Air Conditioning systems, Cooling Towers, DG sets

Unit-V

Energy Economics- discount period, payback period, internal rate of return, net present value; Life Cycle costing ESCO concept.

REFERENCE BOOKS:-

1. Witte L.C. , Schmidt P.S. and Brown D.R., Industrial Energy Management and Utilization, Hemisphere Publ., Washington, 1988.
2. Callaghan P.W., Design and Management for Energy Conservation, Pergamon Press, Oxford, 1981.
3. Murphy W.R. and McKay G., Energy Management, Butterworths, London, 1987.
4. Energy Manager Training Manual , Bureau of Energy Efficiency (BEE) under Ministry of Power, GOI, 2004 (available at www.energymanager training.com).

Bachelor of Technology

Branch - Electrical Engineering

EE-63 Power System Protection

Unit-I:

Introduction and Components of a Protection System Principles of Power System Protection, Relays, Instrument transformers, Circuit Breakers

Unit-II:

Faults and Over-Current Protection Review of Fault Analysis, Sequence Networks. Introduction to Overcurrent Protection and overcurrent relay coordination.

Unit-III:

Equipment Protection Scheme Directional, Distance, Differential protection. Transformer and Generator protection. Bus bar Protection, Bus Bar arrangement schemes.

Unit-IV:

Digital Protection Computer-aided protection, Fourier analysis and estimation of Phasors from DFT. Sampling, aliasing issues.

Unit-V:

Modeling and Simulation of Protection Schemes and System Protection CT/PT modeling and standards, Simulation of transients using Electro-Magnetic Transients (EMT) programs. Relay Testing. Effect of Power Swings on Distance Relaying. System Protection Schemes. Under-frequency, undervoltage and df/dt relays, Out-of-step protection, Synchro-phasors, Phasor Measurement Units and Wide-Area Measurement Systems (WAMS). Application of WAMS for improving protection systems.

REFERENCE BOOKS:-

1. J. L. Blackburn, "Protective Relaying: Principles and Applications", Marcel Dekker, New York, 1987.
2. Y. G. Paithankar and S. R. Bhide, "Fundamentals of power system protection", Prentice Hall, India, 2010.
3. A. G. Phadke and J. S. Thorp, "Computer Relaying for Power Systems", John Wiley & Sons, 1988.
4. A. G. Phadke and J. S. Thorp, "Synchronized Phasor Measurements and their Applications", Springer, 2008.
5. D. Reimert, "Protective Relaying for Power Generation Systems", Taylor and Francis, 2006

EXPERIMENT LIST:-

1. Study of Transformer Protection by Simulation
2. Study of Generator Protection by Simulation
3. Study of Different Characteristics of Over Current Relay
4. Study on Economic Load Dispatch
5. To study the operation of under voltage relay.

Bachelor of Technology

Branch - Electrical Engineering

EE-64 High Voltage Engineering

Unit - I

Breakdown in gases Mechanisms of breakdown in gases, various related ionization processes. Townsends and streamer theories. Paschen's law, Breakdown in Non-uniform fields. Effect of wave shape of impressed voltage on the breakdown strength. Breakdown of sphere gap and rod gap.

Unit - II

Breakdown in liquid and solids Mechanisms of breakdown in liquids, suspended particle, suspended water, cavitation and bubble and electronic breakdown theories. Mechanisms of breakdown in solids; intrinsic electro-mechanical, erosion, surface, thermal and streamer, Relation between electric strength of solids and time, intrinsic breakdown strength.

Unit - III

Impulse Generator Specifications of an impulse voltage Wave, standard impulse, reasons for adopting the particular shape, Analysis and control of simple circuit of impulse generator. Multistage impulse generator (Marx circuit) circuit working, earthing and tripping. Techniques to observe wave front on C.R.O. Generation of High Voltage Methods of generation of power frequency high voltage cascade transformers and resonance methods, Generation of high voltage d.c., voltage stabilization. Tesla coil.

Unit - IV

Measurement of High Voltage Potential dividers-resistive, capacitive and mixed dividers for high voltage. Sphere gap; construction, mounting, effect of nearby earthed objects, humidity and atmospheric conditions, effect of irradiation and polarity, Electrostatic voltmeter; principle and classification, constructional details of an absolute electrostatic voltmeter. Oscilloscopes and their applications in high voltage measurement.

Unit - V

High Voltage Testing Measurement of insulation resistance of cables. Wet and dry flashover test of insulators. Testing of insulators in simulated polluted conditions. Testing of transformers and rotating machines. Measurement of breakdown strength of oil. Basic techniques of non-destructive testing of insulators; measurement of loss angle, High Voltage Schering bridge, and partial discharge measurement techniques

REFERENCE BOOKS:-

- 1 L. V. Bewley, "Traveling Waves on Transmission Systems", Wiley New York.M
- 2 S. Naidu and V. Kamaraju, "High Voltage Engineering", Tata McGraw Hill.
- 3 D.V. Razevigi: "High Voltage Engineering", translated by Dr.M.P.Chourasia,
- 4 Khanna Publisher.

Bachelor of Technology
Branch - Electrical Engineering
EE-66 Minor project-II

Students have to complete minor project by using any software or hardware.
Technically it should be implemented for skill development of the students